

Olga Isupova

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Education

- 2014–** *PhD*
2017 [The University of Sheffield](#), Department of [Automatic Control and Systems Engineering](#), UK
Thesis: *Machine learning methods for behaviour analysis and anomaly detection in video*.
Development of methods for autonomous video scene interpretation and abnormal event detection.
Methodology included topic modeling, Bayesian nonparametrics, change point detection, approximated and sampling inference schemes, image and video processing
Supervisor: [Prof. Lyudmila Mihaylova](#), Professor of Signal Processing and Control.
- 2010–** Finished [Yandex School of Data Analysis](#), Russia
2013 Master level program of [Data Analysis](#)
Study on algorithms and data structures, probabilistic methods in decision making theory, machine learning theory, convex theory and optimisation, linguistics and machine translation and others
- 2007–** *Specialist in Mathematics and System Programming* with Honours, degree equals to Masters
2012 [Lomonosov Moscow State University](#), Department of [Mathematical Methods of Forecasting](#), Russia
Thesis: *Hierarchical topic modeling in document collections by probabilistic latent semantic analysis*.
Building a topic hierarchy tree from an unlabelled document collection with statistical hypothesis testing and topic modeling
Supervisor: [Prof. Konstantin Vorontsov](#), Lecturer at the Lomonosov Moscow State University

Employment History

- 12.2017 - present** - *Postdoctoral Research Assistant in Machine Learning*
[University of Oxford](#)
Machine learning for disaster management and environment protection (within International Partnership Programs in Malaysia and Kenya and Ethiopia). Development of joint learning methods for consolidated answers from crowdsourced labels and convolutional neural networks. Inference of socio-economic impact of droughts
- 04.2017 - 11.2017** - *KTP Research Associate*
[The University of Liverpool](#)
Knowledge transfer partnership [project](#) with [Fletchers Solicitors](#). Development of stand-alone decision making systems on prediction of success of real legal cases based on initial information including structured information, free text clients' stories and evidence records
- 04.2014 - 03.2017** - *Marie Curie Early Stage Researcher at Marie Curie FP7 ITN TRAX project*
[The University of Sheffield](#)
Research topic is "Knowledge Extraction from data". Development of new algorithms for information extraction from complex, heterogeneous data. Main applications: Bayesian methods for anomaly detection and sparse modelling
- 05.2016 - 08.2016** - *Research visits of Automatic Control Group*
[The Linköping University](#)
Joint work with [Prof. Fredrik Gustafsson](#) on change point detection with Gaussian processes

03.2011 - *Research Associate*

02.2014 [Croc Inc](#)

Machine learning research and development: design of data mining algorithms, outliers filtering algorithms; data pre-processing

Business application functional implementation: design of business processes for data analysis, automation of developed algorithms, customer interviewing

06.2011 - *Analyst of the Customer Relationship Management department*

09.2011 Tehnosila – electrical retailer

Data analysis in retail operations: sampling customers for target marketing campaigns; analysis of communication channels, results of marketing actions; statistical analysis of customers' databases

Expertise

■ Maths

Probability theory, Statistics, Stochastic processes
Machine learning, Bayesian nonparametrics, Deep learning
Graphical models, Approximate Inference, MCMC

■ Programming and Miscellaneous

C/C++, Matlab, Python
Tensorflow, Keras
Databases, SQL

Other research related activity

■ Teaching

Laboratory assistant and lab tasks designer for the master course “Intelligent and Vision Systems”, Department of Automatic Control and Systems Engineering, the University of Sheffield, Spring term 2016

■ Talks

Talk “Modern Video Analytics Systems” at the Sheffield Festival of Science and Engineering as part of British Science week 2016, March 2016;

Talk “Abnormal Behaviour Detection in Video” at the PGR symposium at the University of Sheffield, November 2014

■ Served as chairs and organisers

Session chair at the 19th International Conference on Information Fusion, July 2016;
The organising committee of the summer school for the TRAX project, Sheffield, June-July 2015;
Athena Swan Committee at the department of ACSE, the University of Sheffield

■ Served as reviewers

IEEE Transactions of Neural Networks and Learning Systems, Signal Processing Elsevier journal, Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), American Control Conference, IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)

Awards

■ Springer Thesis Series for Outstanding Ph.D. Research

PhD thesis is selected for publishing in the [Springer Thesis Series for Outstanding Ph.D. Research](#)

■ IET Control and Automation Doctoral Dissertation Awards 2017

The second place in the [Institution of Engineering and Technology's \(IET\) Control and Automation Doctoral Dissertation Awards 2017](#)